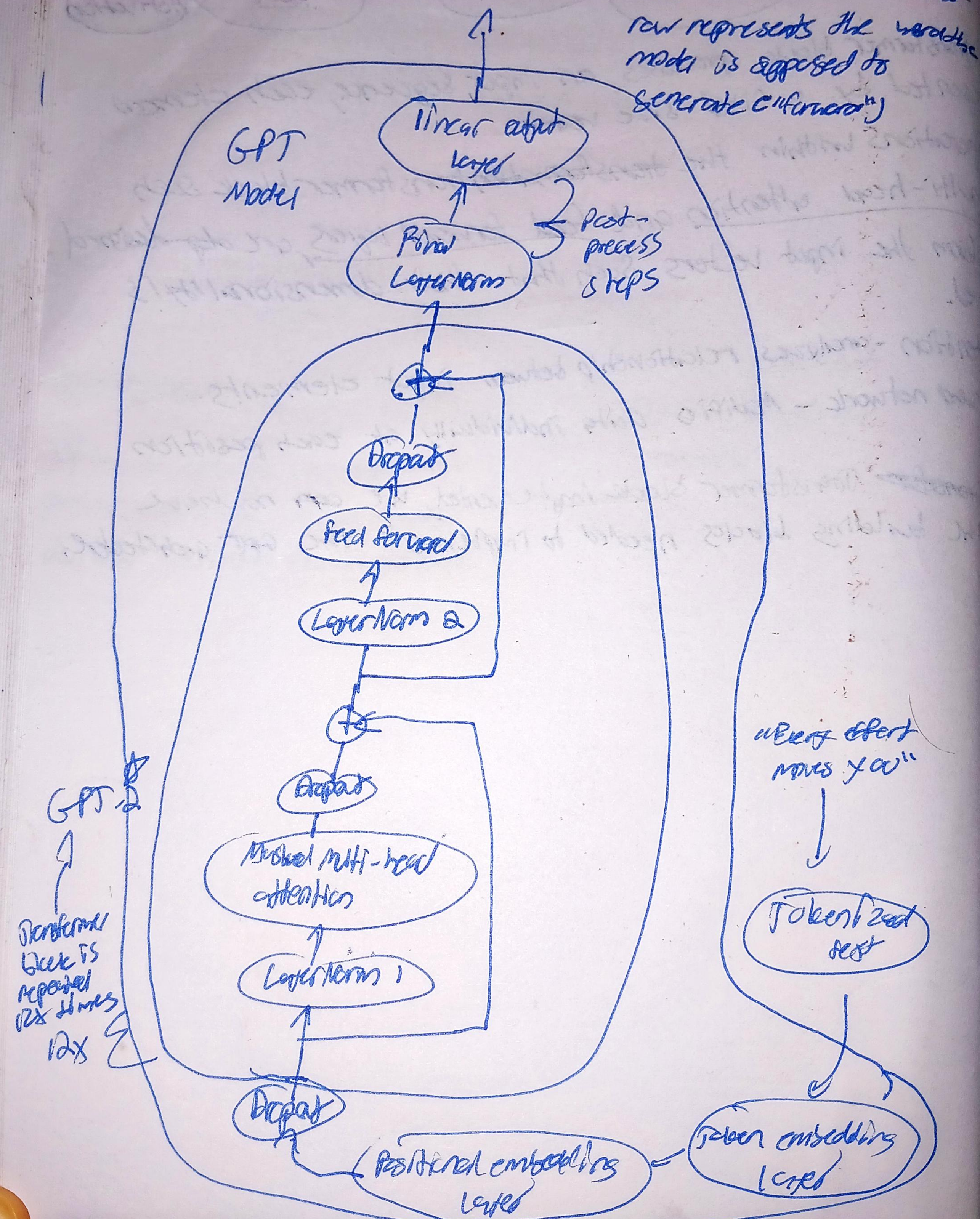
Coding the GPT Model

A 4x50,257-dimensional tensor [[-0.0055, ..., -0.4747], [-0.2663, ..., 0.4224], [1.114, ..., 1.23], [-0.182, ..., -393]]

Goal for these embeddings is to be converted back into text such that the last row represents the word the model is supposed to generate ("forward")



In code, we can now assemble a fully working version of the original ~~$~~124 M parameter version of GPT-2

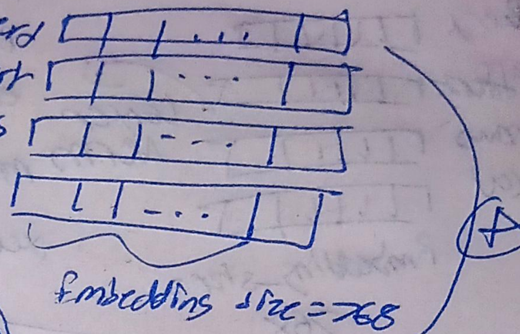
Every effort moves you ~~?~~ Predict

Input Batch:
[[6109, 3626, 6100, 345],
[6109, 1110, 6622, 257]])

Vocab size = 50,257
Every token is mapped to token ID.

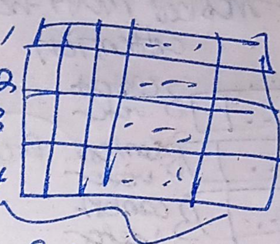
① Token Embeddings:

Every
Effort
Moves
you



② Positional Embedding

~~Every~~ Position 1
Effort Position 2
Position 3
Position 4



③ Input Embeddings = Token embedding + Positional embeddings

Every (pos. 1): Token emb. + Position emb. = Input embed.

Effort (pos 2) " " " "

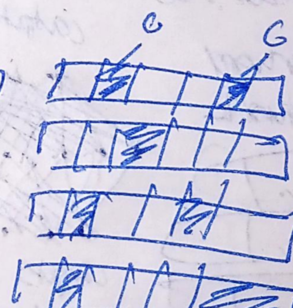
Moves (pos. 3) " " " "

you (pos 4). " " " "

④ Dropout: Input Embeddings

Every
Effort
moves
you

Randomly turn off some elements to 0



⑤ Transformer

μ=mean, σ^2=variance

Input embeddings w/ dropout

Every
Effort
moves
you

Embeddings_size = 768

Layer Norm

Every — $\mu=0$, $\sigma^2=1$
effort — $\mu=0$, $\sigma^2=1$
moves — $\mu=0$, $\sigma^2=1$
you — $\mu=0$, $\sigma^2=1$

Embedding size=768

Masked multi-head attention

Every — context vector
effort — context vector
moves — context vector
you — context vector

emb_size = 768

Dropout

Emb_size=768

Shortcut connections

LayerNorm

Every — $\mu=0$, $\sigma^2=1$
Effort — "
moves — "
you — "

Emb_size=768

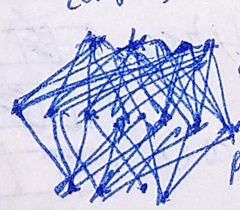
Every
effort
moves
you

Emb_size=768

Feedforward neural network

Every
Effort
Moves
you

Emb_dim=768



Dropout

Emb_size = 768

TURN PAGE

Shortcut connections

μ = mean,
σ^2 = variance

Every
effort
moves
you

Transformer Block Output

Emb_size = 768

6) Layer Normalization

Transformer Block output

Every
effort
moves
you

Emb_size = 768

Layer Norm

Every $\mu=0$, $\sigma^2=1$
effort $\mu=0$, $\sigma^2=1$
moves $\mu=0$, $\sigma^2=1$
you $\mu=0$, $\sigma^2=1$

Emb_size = 768

7) Output head

Every
effort
moves
you

Emb_size = 768

Neural Network Layer (768×50257)

4 prediction tasks

Output Logits for 1 test sample.

Every
effort
moves
you

vocabulary size = 50257

Each element here corresponds to the probability of A being the next token.

2×4×50257
(batch_size, context_size, vocab_size)